

5(a)	1 restriction endonuclease cuts, vector / plasmid ; A restriction enzyme 2 reverse transcriptase to make cDNA using mRNA ; 3 DNA ligase joins sugar phosphate backbone (between gene and vector) or DNA ligase forms phosphodiester bonds (between gene and vector) ;	3
5(b)	<u>marker</u> ; no fluorescence means GFP gene was deleted ; ora	2
5(c)(i)	67.6 or 68 ;	1
5(c)(ii)	higher, as expect 50% (of offspring to get GFP gene from heterozygous male) ; ecf lower, as expect 50%, (if answer to (c)(i) less than 50%)	1
5(c)(iii)	χ^2 / chi-squared ;	1

Question	Answer	Marks
5(c)(iv)	1 10 ng mm⁻³ ; <i>any two from</i> 2 more blastocysts ; ora 3 less toxic ; ora 4 no blastocysts seen under filter / as successful as higher concentrations / all blastocysts have deleted GFP ;	3
5(d)	<i>any three from</i> 1 lanes 1–4 show 4 kbp fragment ; 2 so technique is 100% successful ; 3 (6 kbp gene has) 2 kbp, deleted / lost ; 4 pigs (1–4) have no (normal cell surface) protein ; 5 PRRSV / virus, cannot infect the, cells / pigs (1–4) ;	3

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